

ASSIGNMENT 1

1.

a. Compile it so that it compiles with debugging symbols [using proper option]

Ans:

```
[be2337@localhost assign1]$ gcc a.c -g -c -I.  
[be2337@localhost assign1]$ gcc b.c -g -c -I.  
b.c: In function 'f1':  
b.c:3:3: warning: incompatible implicit declaration of built-in function 'printf' [enabled by default]  
    printf("The numbers are : ");  
    ^  
  
[be2337@localhost assign1]$ gcc a.o b.o -o output  
[be2337@localhost assign1]$ gdb output
```

b. Put breakpoint to function f1.

Ans:

```
lab/Assignments/assign1/output...done.  
(gdb) break f1
```

c. Put breakpoint to line 10 of b.c

Ans:

```
Breakpoint 1 at 0x400756: file b.c, line 3.  
(gdb) break b.c:10
```

d. Run the program until it finishes. Which commands are you using to take it to completion?

Ans:

```

Enter a number between 2 and 6 (non-inclusive):
4
You have entered 4

Breakpoint 1, f1 (x=50, y=163) at b.c:3
3      printf("The numbers are : ");
Missing separate debuginfos, use: debuginfo-install glibc-2.17-157.el7_3.2.x
86_64
(gdb) c
Continuing.
The numbers are : < 50, 163>

Breakpoint 2, f2 (p=0x7fffffff2c4, q=0x7fffffff2c0)
    at b.c:10
10      }
(gdb) c
Continuing.

Breakpoint 1, f1 (x=163, y=50) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
After operation 1 The numbers are : < 163, 50>

Breakpoint 1, f1 (x=33, y=109) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
The numbers are : < 33, 109>

Breakpoint 2, f2 (p=0x7fffffff2c4, q=0x7fffffff2c0)
    at b.c:10
10      }
(gdb) c
Continuing.

Breakpoint 1, f1 (x=109, y=33) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.

```

```

After operation 2 The numbers are : < 109, 33>

Breakpoint 1, f1 (x=25, y=81) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
The numbers are : < 25, 81>

Breakpoint 2, f2 (p=0x7fffffff2c4, q=0x7fffffff2c0)
    at b.c:10
10      }
(gdb) c
Continuing.

Breakpoint 1, f1 (x=81, y=25) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
After operation 3 The numbers are : < 81, 25>

Breakpoint 1, f1 (x=20, y=65) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
The numbers are : < 20, 65>

Breakpoint 2, f2 (p=0x7fffffff2c4, q=0x7fffffff2c0)
    at b.c:10
10      }
(gdb) c
Continuing.

Breakpoint 1, f1 (x=65, y=20) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
After operation 4 The numbers are : < 65, 20>
[Inferior 1 (process 15829) exited with code 04]

```

e. How many times breakpoint “1” is hit in one run of the program ?

Ans:

```
(gdb) info break 1
Num      Type           Disp Enb Address            What
1        breakpoint      keep y   0x0000000000400756 in f1 at b.c:3
          breakpoint already hit 8 times
```

f. How many times breakpoint “2” is hit in one run of the program

Ans:

```
(gdb) info break 2
Num      Type           Disp Enb Address            What
2        breakpoint      keep y   0x00000000004007c6 in f2 at b.c:10
          breakpoint already hit 4 times
```

g. How you can see details about a breakpoint ?

Ans: Using info break

h How you can see details about all breakpoints ?

Ans:

```
(gdb) info breakpoints
Num      Type           Disp Enb Address            What
1        breakpoint      keep y   0x0000000000400756 in f1 at b.c:3
          breakpoint already hit 8 times
2        breakpoint      keep y   0x00000000004007c6 in f2 at b.c:10
          breakpoint already hit 4 times
```

i. What is value of variable x in f1 when breakpoint “1” is hit for 3 rd time ? How you can examine it

Ans: X=33

```
Enter a number between 2 and 6 (non-inclusive):
4
You have entered 4

Breakpoint 1, f1 (x=50, y=163) at b.c:3
3      printf("The numbers are : ");
Missing separate debuginfos, use: debuginfo-install glibc-2.17-157.el7_3.2.x
86_64
(gdb) c
Continuing.
The numbers are : < 50, 163>

Breakpoint 2, f2 (p=0x7fffffff2c4, q=0x7fffffff2c0)
at b.c:10
10     }
(gdb) c
Continuing.

Breakpoint 1, f1 (x=163, y=50) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
After operation 1 The numbers are : < 163, 50>

Breakpoint 1, f1 (x=33, y=109) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
The numbers are : < 33, 109>
```

J. Rerun the program.put a breakpoint at function f0. list 5 lines where it has stopped with breakpoint 3 for firsttime.

```
Reading symbols from /home/usr/student/ug/yr23/be2337/selab/Assignments/assign1/output...done
(gdb) break f1
Breakpoint 1 at 0x400756: file b.c, line 3.
(gdb) break b.c:10
Breakpoint 2 at 0x4007c6: file b.c, line 10.
(gdb) break f0
Breakpoint 3 at 0x4005f9: file a.c, line 6.
(gdb) run
Starting program: /home/usr/student/ug/yr23/be2337/selab/Assignments/assign1/output

Breakpoint 3, f0 (p=0x7fffffff2c8) at a.c:6
6      int x, cntr = 1;
Missing separate debuginfos, use: debuginfo-install glibc-2.17-157.el7_3.2.x86_64
(gdb) set listsize 5
(gdb) 1
Undefined command: "1". Try "help".
(gdb) 1
1
2
3      #include <f.h>
4
5      int f0(int *p) {
(gdb) c
Continuing.
Enter a number between 2 and 6 (non-inclusive):
4
You have entered 4

Breakpoint 1, f1 (x=50, y=163) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
The numbers are : < 50, 163>

Breakpoint 2, f2 (p=0x7fffffff2c4, q=0x7fffffff2c0) at b.c:10
10     }
(gdb) c
Continuing.
```

Continuing.

Breakpoint 1, f1 (x=163, y=50) at b.c:3

```
3      printf("The numbers are : ");
```

```
(gdb) c
```

Continuing.

After operation 1 The numbers are : < 163, 50>

Breakpoint 1, f1 (x=33, y=109) at b.c:3

```
3      printf("The numbers are : ");
```

```
(gdb) c
```

Continuing.

The numbers are : < 33, 109>

Breakpoint 2, f2 (p=0x7fffffff2c4, q=0x7fffffff2c0) at b.c:10

```
10     }
```

```
(gdb) c
```

Continuing.

Breakpoint 1, f1 (x=109, y=33) at b.c:3

```
3      printf("The numbers are : ");
```

```
(gdb) c
```

Continuing.

After operation 2 The numbers are : < 109, 33>

Breakpoint 1, f1 (x=25, y=81) at b.c:3

```
3      printf("The numbers are : ");
```

```
(gdb) c
```

Continuing.

The numbers are : < 25, 81>

Breakpoint 2, f2 (p=0x7fffffff2c4, q=0x7fffffff2c0) at b.c:10

```
10     }
```

```
(gdb) c
```

Continuing.

Breakpoint 1, f1 (x=81, y=25) at b.c:3

```
3      printf("The numbers are : ");
```

```
(gdb) c
```

Continuing.

After operation 3 The numbers are : < 81, 25>

```

Breakpoint 1, f1 (x=20, y=65) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
The numbers are : < 20, 65>

Breakpoint 2, f2 (p=0x7fffffffef2c4, q=0x7fffffffef2c0) at b.c:10
10     }
(gdb) c
Continuing.

Breakpoint 1, f1 (x=65, y=20) at b.c:3
3      printf("The numbers are : ");
(gdb) c
Continuing.
After operation 4 The numbers are : < 65, 20>
[Inferior 1 (process 22658) exited with code 04]

```

EXPLORE:

```

(gdb) info breakpoints
Num      Type             Disp Enb Address              What
1        breakpoint      keep y   0x0000000000400756 in f1 at b.c:3
         breakpoint already hit 8 times
2        breakpoint      keep y   0x00000000004007c6 in f2 at b.c:10
         breakpoint already hit 4 times
3        breakpoint      keep y   0x00000000004005f9 in f0 at a.c:6
         breakpoint already hit 1 time
(gdb)

```